



Implementation Tools: Using Data Transfer Workbench (DTW)

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the course on using the Data Transfer Workbench (DTW).
This course should be useful to anyone using DTW for the first time.

Objectives



On completion of this course you will be able to:

- Describe the process for importing data using Data Transfer Workbench
- List the type of objects that can be imported using DTW
- Explain the relationship between SAP Business One objects and the provided DTW templates
- Explain how to improve performance when importing into a SAP HANA database
- Explain how to update child records for an object
- Describe how to create a new template by customizing an existing template

In this topic, you will see the process and steps for using Data Transfer Workbench to import objects. You will also be able to:

- List the type of objects that can be imported using DTW
- Explain the relationship between SAP Business One objects and the provided DTW templates
- Explain how to improve performance when importing into a SAP HANA database
- Describe how to create a new template by customizing an existing template

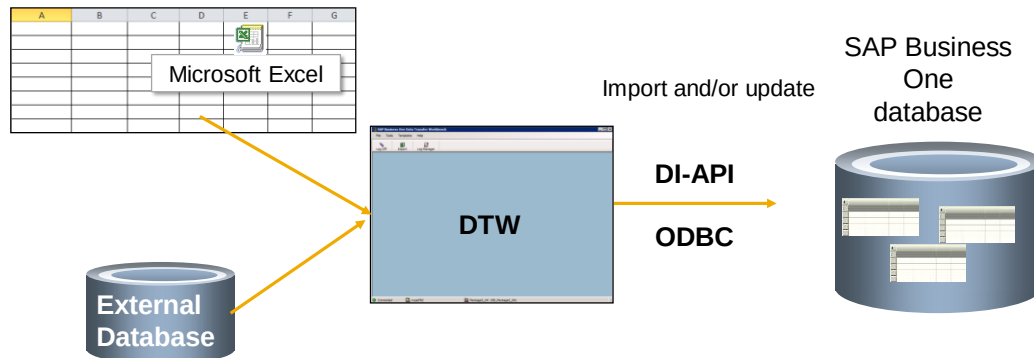
Overview and References



In this segment we introduce the Data Transfer Workbench tool and provide some references to help you with the objects and fields.

Data Transfer Workbench (DTW)

- Data Transfer Workbench (DTW) is a tool provided with SAP Business One for bulk import of data into SAP Business One objects using predefined Microsoft Excel templates
- DTW supports the import of master data, setup data, journal entries and open sales, purchasing production and service documents.



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Data Transfer Workbench (DTW) is a tool provided with SAP Business One for importing data into SAP Business One objects using predefined Microsoft Excel templates. The Data Transfer Workbench is designed primarily for bulk import or update of master data, setup data, chart of accounts, journal entries and open sales, purchasing, production and service documents.

The financial data you can import includes opening balances, journal entries and journal vouchers.

You can also import setup data, such as users, tax codes, and banks.

DTW uses the DI API to access the SAP Business One database. Some familiarity with SAP Business One table and field names is required. For this reason, end users might be more comfortable with using the Import from Excel utility as a periodic import tool.

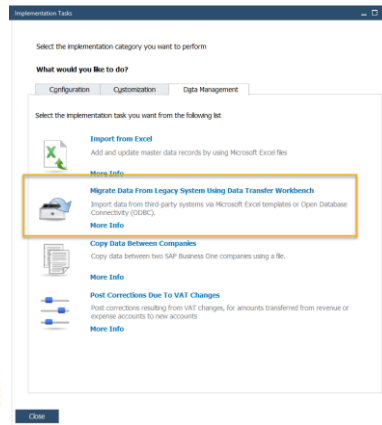
DTW can also import data from external databases, using SQL queries over an ODBC connection.

Note: The Data Transfer Workbench does **not** support the import of documents that have already been closed. It also does **not** support the deletion of records from the database.

Starting DTW

Start from desktop:
Program Files > SAP > Data Transfer Workbench > DTW.exe

Start from SAP Business One implementation center: *Administration > System Initialization > Implementation Center > Implementation Tasks*



To start DTW from the implementation center you must provide the path to the DTW executable in the *Settings* window



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Data Transfer Workbench (DTW) is supplied with the SAP Business One client installation files and is an optional package.

The DTW package can be installed on the client desktop or on a remote desktop without the SAP Business One client. Since DTW uses the DI API for access to the database, then the DI API should also be selected for installation on the same desktop.

To start DTW from the desktop, open the executable DTW.exe.

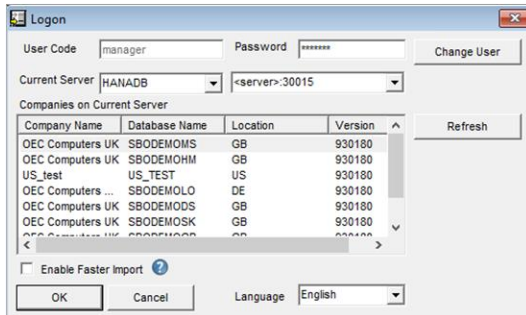
You can also start DTW from the Implementation Center within SAP Business One. To do this, select the link on the Data Management tab of the Implementation Tasks window.

If you wish to start DTW from the Implementation Tasks window you must provide the path to the DTW executable in the Implementation Center *Path Settings* window.
Note: Microsoft Excel should be installed on the system where DTW is installed.

Connecting to the Database Server

In the DTW logon window, you need to login to SAP Business One target company as a super user

Servers are displayed from SLD



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In the DTW Login window:

- The database server type and server names are taken from the System Landscape Directory (SLD). Only registered servers and company databases are displayed. You no longer need to login to the database server.

Note for SAP HANA the port number (for example 30015) is required in addition to the server name or IP address.

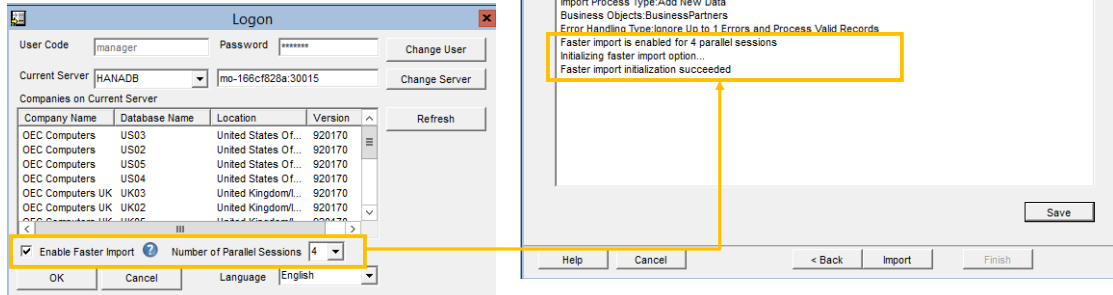
You should then see one or more company databases that exists on the chosen server. If you do not see a company displayed, choose the *Refresh* button to reload the list of companies on the selected server.

Select the relevant SAP Business One company and login with a super user account. A super user is required since this user must have full authorization to objects.

The *Enable Faster Import* checkbox in the Logon window allows for faster import of records into SAP HANA databases by running the import using multiple threads. This will be covered later in this course.

Enable Faster Import for SAP HANA

- With an SAP HANA database, you can run the import using multiple, parallel processes
- Select the *Enable Faster Import* checkbox, and select the number of processes (default is 4)



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When you use a SAP HANA database you can increase the performance of the import by running multiple parallel DI API processes.

You must make the selection when you logon to DTW, by checking the *Enable Faster Import* checkbox. Then select the number of processes. The default is 4 but you can select up to 21 threads, depending on the hardware configuration size you have.

When you select *Enable Faster Import*, DTW will run the DI API under a COM+ application. The rows from the spreadsheet will be shared out to the threads and imported in parallel. Thus the order in which the records are imported can differ from the order of the spreadsheet.

There are some limitations on the use of this option:

- The performance increase is only effective for imports of more than 1000 records.
- You cannot run a simulation before the import (the button is deactivated).
- Only certain objects are supported: Setup data is not supported at all. For master data, only business partner and item master data are supported. For documents, only quotations, orders, deliveries, goods receipts and invoices are supported.

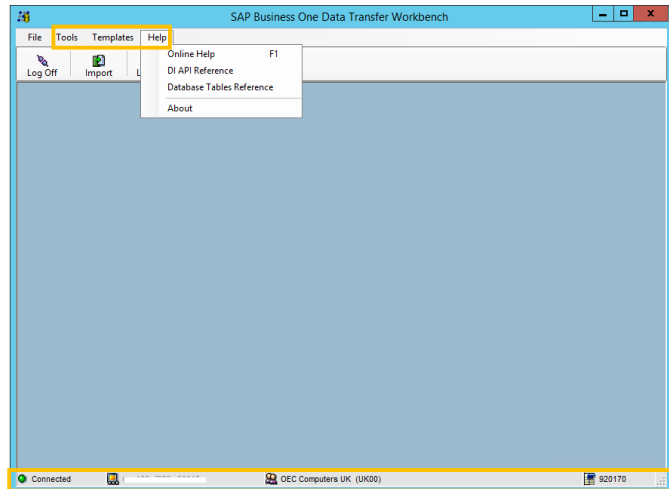
See SAP note 2229208 for more information.

DTW Menu and Help

After logging in, the DTW application window opens showing the menu options:

- Tools: Import wizard, log manager
- Templates: creation of new templates
- Help: DTW online help, DI API Reference and Database Tables Reference

The bottom of the application window shows the status of the connection to the target database, including the company



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After you successfully login to the company, the DTW application window opens:

- To start the import wizard, use the *Tools* menu or choose the *Import* button. From the *Tools* menu you can also access log files from previous import runs.
- Use the *Templates* menu to generate new templates. This functionality is covered later in this course.
- Under the *Help* menu you can find the online help, the DI API Reference and the Database Tables Reference. You may need to reference these resources when entering data into the templates:
 - The DI API Reference describes objects and object properties
 - The Database Tables Reference shows the table fields including default values for fields, field types, and constraints on field values

At the bottom of the DTW application window you can see the status of the connection to the target database, including the server and target company name.

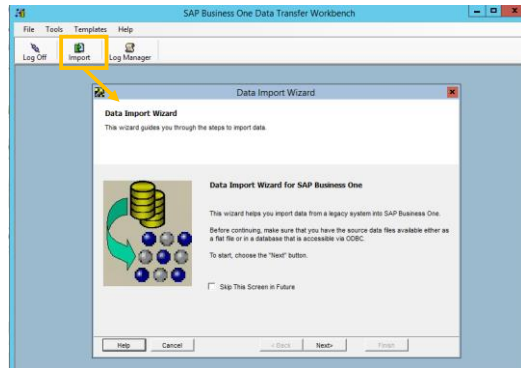
Overall Process for Import

Enter data into predefined templates

Save template as delimited file*

Log in to target company database

In DTW, choose *Import* button to run import wizard



*tab delimited, comma delimited, or semi-colon delimited file.

Note: the delimiter character should not appear in any field data.

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To import data, you will use the predefined templates and enter your data in the columns of each template.

After you have entered the data, save the template as a tab, comma, or semi-colon delimited file type. Any of these file types is accepted by DTW.

In DTW, log in to the target company as a super user, and start the import wizard.

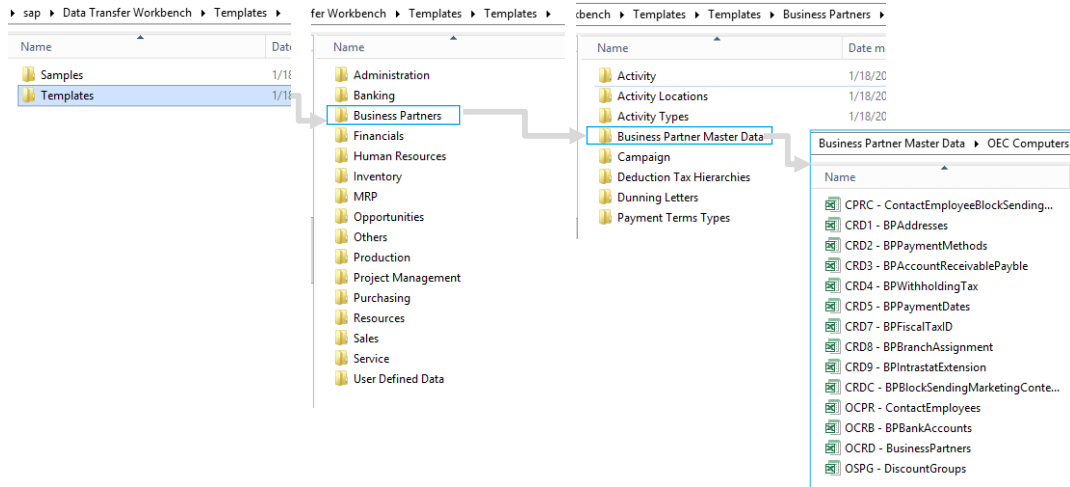
Note: The delimiter character (tab, comma or semi-colon) should **not** be present in any of the data in the template, since it will be interpreted as the end of the data. Thus, if the data is likely to contain commas or semi-colons, save as a tab delimited file.

Predefined Templates



Let's look at the predefined templates provided in DTW.

DTW Templates



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After DTW is installed, the DTW executable folder contains the predefined Microsoft Excel templates (in the Templates subfolder). Some sample templates are also provided in this folder.

The templates are organized by folders similar to the main menu in the SAP Business One client.

Predefined Microsoft Excel templates are provided for each object that can be imported. An object in SAP Business One can be represented by multiple tables in the database, and each predefined template aligns to a table for the object. The template names start with the four character database table name, so, for instance, the parent business partner master data template is OCRD – *BusinessPartners*. A template is provided for each of the child tables. For the business partner master data object the child templates include CRD1, CRD2, OCPR, OCRB, and so on.

The Samples folder contains examples of completed templates.

Example: OCRD Template

- Each predefined template includes a column for each field exposed to the DI API
- The two header rows should not be removed. They contain the object properties and the DI API field names

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	CardCode	CardName	CardType	GroupCode	Address	ZipCode	MailAddress	MailZipCod	Phone1	Phone2	Fax	ContactPei	Notes	PayTerms
2	CardCode	CardName	CardType	GroupCode	Address	ZipCode	MailAddress	MailZipCod	Phone1	Phone2	Fax	CntctPrsn	Notes	GroupNum
3														
4														
5														

- These fields are described in the *DI API Objects Reference* and the *Database Tables Reference*
- SAP recommends you do not remove columns from a template; instead leave a column blank, or generate a customized template

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The predefined templates include a column for every field in the table that is exposed to the DI API. Each template contains two header rows which should not be removed:

- The names in the first header row are the object properties. These are documented in the DI API – Objects Reference.
- The names in the second header row are the DI API field names. These are documented in the Database Tables Reference

Sometimes the names are identical.

If you do not want to import data for a field, or you want to use the default value, leave the column blank.

SAP recommends you do not remove columns from the original template. If you want to remove columns, you can generate a customized template from an existing template. Customized templates will be covered later in this course.

Field Comments and Enumerated Data

- Each header field in the template includes a comment showing data type and field length
- If the type is shown as enum, then you must use one of the valid values shown
- If the comment shows a related table, then enter the code of the related table in the template field

A	B	C	D	E	F	G	L	M	N	O	P	Q	R	S	T	U	V
1	CardCode	CardName	CardType	GroupCode	Address	ZipCode	MailAddress	ContactPerson	Notes	PayTerms	CreditLimit	MaxComm	DiscountP	VatLiab	FederalTax	Deductible	Deduction
2	CardCode	CardName	CardType	GroupCode	Address	ZipCode	MailAddress	ContactPerson	Notes	GroupNum	CreditLine	DebitLine	Discount	VatStatus	LicTradNum	DdctStatus	DdctPrnt
3																	

Description: BP Code
 Type: string
 Field Length: 15
 This field is mandatory

Description: Group Code
 Type: long
 Field Length: 6
 Related Table: OCRG

Description: Tax Definition
 Type: enum
 Valid Values: vExempted,vLiab,vEC

Query Preview

SELECT * FROM OCRG T0

#	Group No.	Group Name	Group Type	U
1	100	Customers	C	N
2	101	Suppliers	S	N
3	102	Large Accounts	C	N
4	103	Distributors	C	N
5	104	Small Accounts	C	N
6	105	Government	C	N
7	106	High Tech	C	N

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Each field in the template heading row includes a comment. The comment pops up when you hold the mouse over the field and shows the field description, type, and length.

If the field is mandatory, such as a code, this is indicated in the comment.

If the type shows as enum, then DTW expects you to enter one of the valid values shown. These are the values expected by the DI API. In the example, in the business partner master data template, the CardType (column C) and VatLiab (column R) fields have enumerated data. As an example, in the CardType field you must enter cCustomer, cSupplier or cLid and not the values as specified in the Database Tables Reference.

If the comment shows the field with a related table, this column holds the key to the related table, and in this column you enter the *key value*. For example, in the GroupCode field (column D), you would enter the code allocated by the system when you create new customer or supplier groups. To see the codes run a query on the table.

Valid Values and Database Tables Reference

- If a field has valid values in the template comment, always use these values and do not enter the constraints shown in the Database Tables Reference

The screenshot illustrates the relationship between the SAP Business One Database Tables Reference and the Excel spreadsheet template. The SAP window shows the 'Business Partner' table with fields like CardCode, CardName, CardType, GroupCode, CmpPrivate, Address, and ZipCode. The Excel spreadsheet shows a template for 'BusinessPartners1' with columns A through I. A comment box in the Excel spreadsheet for the CardType field (column C) contains the text 'Valid Values: cCustomer,cSupplier,cLid'. An arrow points from the 'CardType' field in the SAP window to the comment box in the Excel spreadsheet.

If the valid values shown in the comment are different to the values documented in the Database Tables Reference, you *must always* enter one of the values in the comment, and not use the constraints from the Database Tables Reference.

As an example, the Database Tables Reference shows the constraints for the CardType field (column C) as C, S and L (customer, vendor and lead). However, DTW expects one of the values cCustomer, cSupplier or cLid.

You can however leave the field blank if the CardType is a customer, since this is the default as indicated in the Database tables Reference.

Leaving Columns Blank

- The data entry rules for DTW are identical to manual data entry. You do not need to enter default field values or non-required data
- Make sure that prerequisite data is created before you import the data

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
CardCode	CardName	CardType	GroupCode	Address	ZipCode	MailAddress	MailZipCod	Phone1	Phone2	ContactPe	PayTerms	CreditLimit	Discount	ValStatus	LicTradNum	FederalTax		
Cust01	Best Hotels	cCustomer	100	5 The Mall	W1 2DW	Brackley Bridge House	E1 2DW	0800-665-2239										
Cust10	Jade Holdings	cCustomer	104	21 East Gate St	EC1 2SA	21 East Gate St	EC1 2SA	021-214-4439										
Cust20	Fancy Foods	cCustomer	104	75 York Rd	LS1 6HG	77 York Rd	LS1 6HG	201-554-9080										
Cust30	DWB Enterprises	cCustomer	106	Rosenthaler Strasse 30	10208	Rosenthaler Strasse 30	10208	030 41052500										
Cust40	Mega Construction	cCustomer	107	Unit 25	W2 6OP	Unit 25	BB3 5WE	0213-3096658										
Supp01	Sound Systems	cSupplier	101	66 New St	B3 7TR	66 New St	B3 7TR	021-408-4908										
Supp10	Spotlights	cSupplier	101	12 Haywood Rd	BB3 5WE	12 Haywood Rd	SN29 2DF	01283-981246										
Supp20	Laser Effects	cSupplier	101	9 Bridge House	OX7 8UO	9 Bridge House	OX7 8UO	01865-352902										
Supp30	Echo F																	
Supp40	Cable																	

AC	AE	AG	AH	AI	AJ	AK	AL	AM	AO	AP	AQ	BW	CA	CB	CC	CD
Currency	Cellular	City	County	Country	MailCity	MailCounty	MailCountry	E-Mail	DefaultAcc	DefaultBra	DefaultBan	PymntM	BankCode	HouseBank	HouseBnkCry	HouseBnkAct
GBP		London		GB	London			info@besthotelg.c	33445678	201016	HSBC		GB	HSBC	GB	23456789
GBP		London		GB	London			sam@jadeh.co.uk	44552100	201045	HSBC		GB	HSBC	GB	23456789
GBP		Leeds	Yorkshire	GB	Leeds	Yorkshire		wm@floods.co.uk	55671234	401087	Barclays		GB	HSBC	GB	23456789
EUR		Berlin		DE	Berlin			dwb@dw bent.com	66789012	102030	10000000		DE	HSBC	GB	23456789
GBP		Oldham	Lancs	GB	Oldham	Lancs			77890123	201035	Barclays		GB	HSBC	GB	23456789
GBP		Bath	Wiltshire	GB	Bath	Wiltshire			11234590	401020	HSBC		GB	HSBC	GB	23456789
GBP		Swindon	Wiltshire	GB	Swindon	Wiltshire			11234521	401030	Barclays		GB	HSBC	GB	23456789
GBP		Oxford	Oxford	GB	Oxford	Oxford			11234532	401045	HSBC		GB	HSBC	GB	23456789
USD		New York		US	New York				11234545	23006900	BNY		US	HSBC	GB	23456789
GBP		Slough	Berkshire	GB	Slough	Berkshire			11234566	401080	Barclays		GB	HSBC	GB	23456789

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When you manually create master data or documents in the SAP Business One application, you take advantage of default field values, and only enter the required or mandatory fields.

The same is true for DTW. You can leave columns blank in the template and enter just the required fields and non-default values.

Shown is an example of a completed OCRD template for the business partner master data header, with columns left blank if there are default values.

Make sure that any prerequisite data is created for the data you wish to import.

For the business partner master data object, prerequisite data includes:

- Business partner groups (GroupCode column D in the template)
- Currencies (Currency column AC)
- Bank codes for business partner banks (BankCode column AQ)
- House bank definitions (columns CB to CD)

Parent and Child Templates

- If there are child tables for an object, you can import the child records together with the parent records
- To relate the child and parent templates, the parent key field in the child template will link back to the primary key (column A) in the parent template.

Parent template

CardCode	CardName	CardType	GroupCode	Address	ZipCode	MailAddress	MailZipCod	Phone1	Phone2	Fax	ContactPe	Notes	PayTermst
C001													

Child template

ParentKey	LineNum	CardCode	Name	Position	Address	Phone1	Phone2	MobilePho	Fax	E-Mail	Pager	Remarks1	Remarks2
C001													

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If there are child tables for an object, such as the contact employees table OCPR for the business partner object as shown here, you can import the child records in the same import run as the parent records.

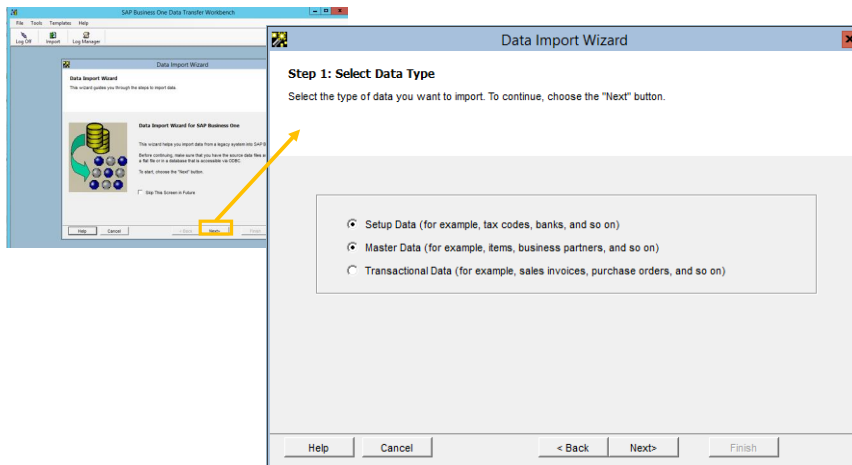
To relate the child and parent templates, the parent key field in the child template will link back to the primary key (column A) in the parent template.

Import Process



Let's now look at the overall process for importing the data.

DTW Import Wizard



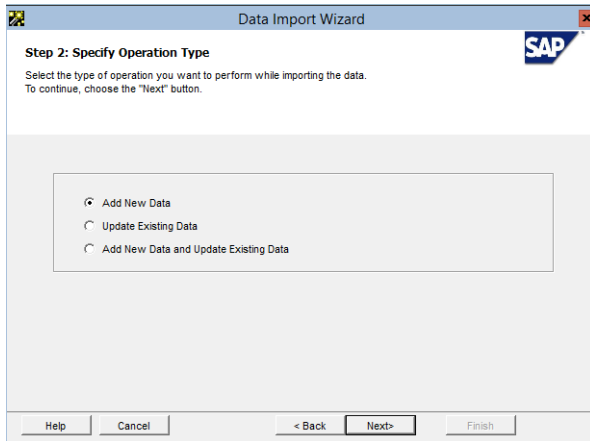
Select the type of data to be imported:

- Setup data
- Master data
- Transactional data

In the first step of the import wizard, select the type of data to be imported (setup data, master data or transactional data):

- Setup data includes banks, house banks, business partner and item groups, tax codes, opening balance data, and users. For the full list of templates, check the subfolders for the Templates folder
- Master data includes business partner, item and employee master data
- Transactional data includes journal entries, sales and purchasing documents, business partner activities, payments, and inventory transactions.

DTW Import Wizard



Step 2: Specify Operation Type

Select the type of operation you want to perform while importing the data.
To continue, choose the "Next" button.

☒ Add New Data
☐ Update Existing Data
☐ Add New Data and Update Existing Data

Help Cancel < Back Next> Finish

Select the operation to be performed:

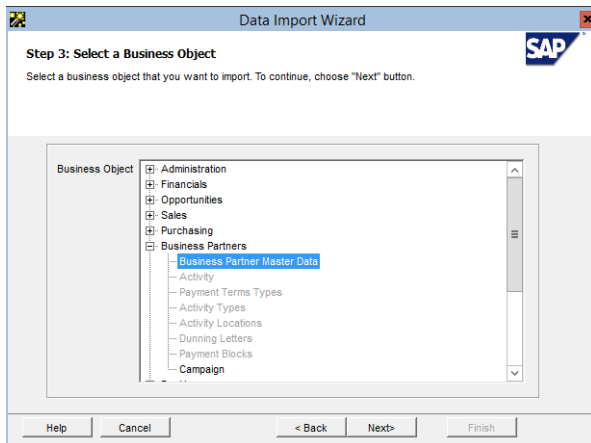
- Add
- Update
- Add and Update

In the next step, select whether you want to import new data (insert) or update data (update):

- To import new data objects, choose "Add New Data."
- To update information in a parent or child template, choose "Update Existing Data."
- To import new parent records for an object while at the same time updating existing child records, choose "Add New Data and Update Existing Data."

Note the selection "Add new data and update existing data" will create a new record in the database if the key does not exist, and update an existing record if the key matches. This selection is not available if transactional data was selected in step 1.

DTW Import Wizard



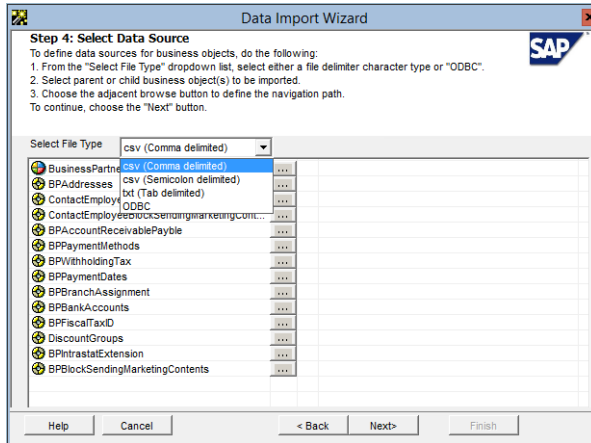
Navigate to the business object by expanding the list

You can only import data for a single business object for each import run

Navigate to the business object by expanding the list. Only objects for the type of data selected in step 1 (e.g., master data) are selectable. The rest are greyed out.

Using the wizard you can only import data for a single business object at a time. For example, you cannot mix business partner master data and item master data in the same import run.

DTW Import Wizard



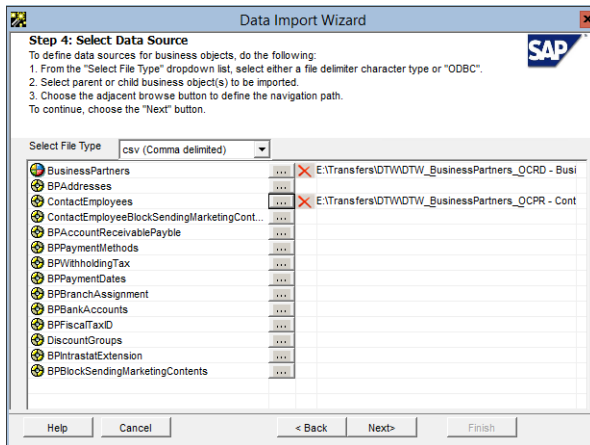
Select the file type for the saved template:

- Comma separated
- Semicolon separated
- Tab delimited
- ODBC

In this step select the format of the saved template files (comma separated, semicolon separated, or tab delimited).

If you select ODBC, you will be prompted to supply a SQL query. The query will extract the data from a source database, and import the data into the target SAP Business One database.

DTW Import Wizard



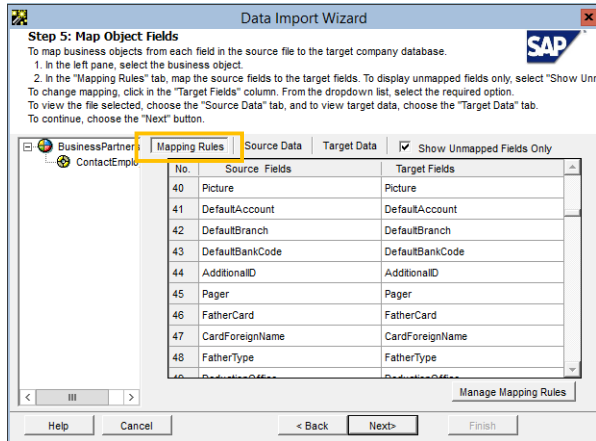
- The wizard displays all the available templates for the selected business object
- For each table to be imported, select the browse button and locate the saved template file (comma, semi-colon or tab delimited)
- After selection, the path to the template is shown

The wizard displays all the available templates for the selected business object.

For each table you want to import, select the browse button and then locate the relevant saved template (saved as a comma, semi-colon or tab delimited file).

A red x indicates a template has been selected and the path is shown.

DTW Import Wizard

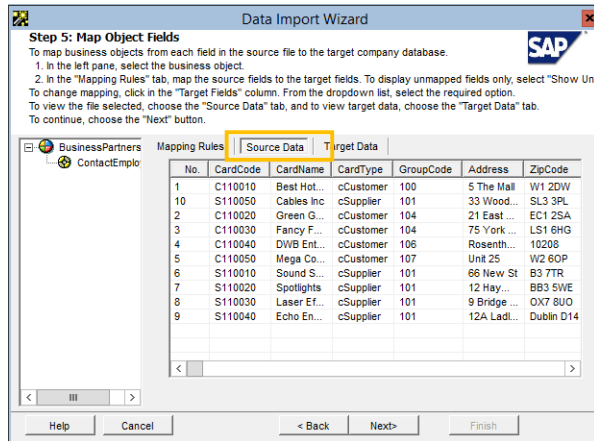


Use the *Mapping Rules* tab to visually check the mapping between the template source data fields and the DI API fields

In this step, using the *Mapping Rules* tab, you can visually check that the fields in the source data template map to the DI API database fields.

If the target field is blank, you can manually map it to the source field. Note: if several target fields are blank, you may have selected the wrong template file.

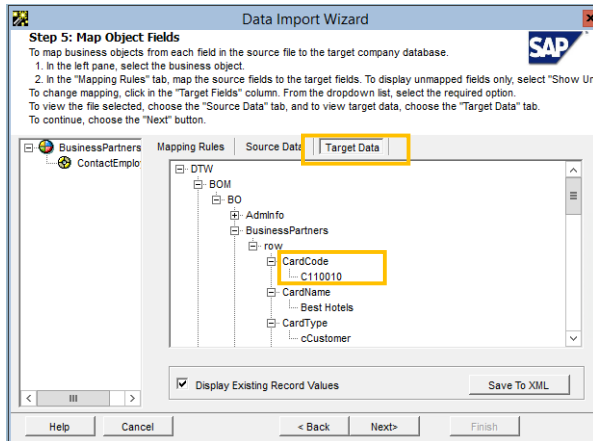
DTW Import Wizard



Use the *Source Data* tab to visually scan the template data to make sure each field is interpreted correctly

On the Source Data tab you can visually scan the input data from the template to make sure the fields are interpreted correctly by DTW.

DTW Import Wizard

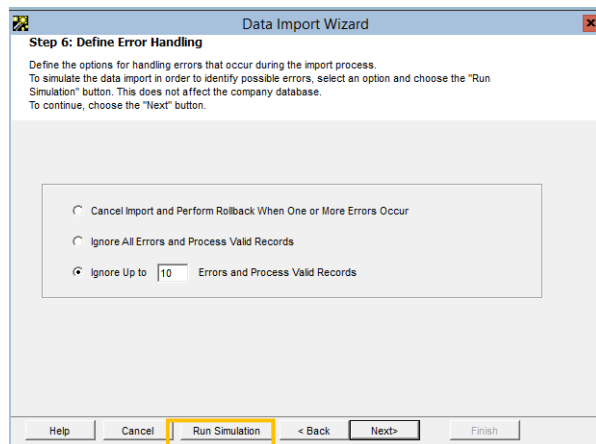


Use the *Target Data* tab to visually scan the expected target data fields based on the mapping

Expand the object structure to see the expected data for each field

Use the *Target Data* tab to visually scan the target data fields based on the mapping. Expand the object (BOM/BO) structure then expand each row and each field to see the expected data. This can be useful for troubleshooting errors during an import. If the target fields are not as expected this might be the cause of the error.

DTW Import Wizard



You can define how the wizard will handle any errors during the import:

- Cancel the import after the first error is encountered
- Ignore all errors and import valid records
- Ignore a set number of errors and import valid records

Option to run a simulation before actual import of records

You can define how the wizard will handle any errors during the import:

- Cancel the import after the first error is encountered
- Ignore all errors and import valid records
- Ignore a set number of errors and import valid records

Also on this step you have the option to run a simulation before the actual import. The simulation will point out any errors but will not import any data. SAP recommends that you run a simulation first, or import the data into a test database so you can validate the results.

DTW Import Wizard

Data Import Wizard

Step 7: Start Importing Data

To start importing data into SAP Business One, choose the "Import" button.
Note that the duration of the data import depends on the number of records imported.
You can save these wizard setting to XML file and run it later in batch mode as a Scheduled Run - press the "Save" button

Import Data Type: Master Data
Import Process Type: Add New Data
Business Objects: BusinessPartners, ContactEmployees
Error Handling Type: Cancel Import and Perform Rollback When One or More Errors Occur

Help Cancel < Back **Import** Finish

Log entry produced for each record imported into the database

Detailed Log 000000000029

Date-Time: 04/19/2017 22:47:26 Company: UK00
Username: manager Log: 000000000029
Server: mo-166c828a-30015 Error File:
Display: All

Loading Summary
Created: 9
Updated: 1
Re-Import Errors

No	Status	Key	Help	Description
1	Created Successfully	C110010		C110010oBusinessPartners
2	Created Successfully	C110020		C110020oBusinessPartners
3	Updated Successfully	C110030		C110030oBusinessPartners
4	Created Successfully	C110040		C110040oBusinessPartners
5	Created Successfully	C110050		C110050oBusinessPartners
6	Created Successfully	S110010		S110010oBusinessPartners
7	Created Successfully	S110020		S110020oBusinessPartners
8	Created Successfully	S110030		S110030oBusinessPartners
9	Created Successfully	S110040		S110040oBusinessPartners
10	Created Successfully	S110050		S110050oBusinessPartners

The final step is the import of data. A log is produced for each record imported successfully into the database.

DTW Errors

Detailed Log 000000000030

Date-Time: 04/19/2017 23:56:00 Company: UK00
 Username: manager Log: 000000000030
 Server: mo-166c828a30015 Error File: C:\Program Files (x86)\ISA

Display: All

Loading Summary: Created: 6, Updated: 0

Buttons: Close, Error File, Re-Import Errors

No	Status	Key	Help	Description
1	Created Successfully	I-56010		I-56010Items
2	Created Successfully	I-56020		I-56020Items
3	Created Failed	I-56030		[OTM.VatGoupSa], 'Linked value 2 does not exist' [OTM.VatGoupSa], 'Linked value 2 does not e...
4	Created Failed	I-56040		[OTM.VatGoupSa], 'Linked value 3 does not exist' [OTM.VatGoupSa], 'Linked value 3 does not e...
5	Created Failed	I-56050		[OTM.VatGoupSa], 'Linked value 4 does not exist' [OTM.VatGoupSa], 'Linked value 4 does not e...
6	Created Failed	I-56060		[OTM.VatGoupSa], 'Linked value 5 does not exist' [OTM.VatGoupSa], 'Linked value 5 does not e...
7	Created Failed	I-56070		Minimum inventory must be less than the required inventory [OTM.MinLevel], 'I-56070' Minimum inven...
8	Created Failed	I-56080		Minimum inventory must be less than the required inventory [OTM.MinLevel], 'I-56080' Minimum inven...
9	Created Successfully	I-581010		I-581010Items
10	Created Successfully	I-581020		I-581020Items
11	Created Successfully	I-581030		I-581030Items
12	Created Successfully	I-581040		I-581040Items
13	Created Failed	I-581050		Minimum inventory must be less than the required inventory [OTM.MinLevel], 'I-581050' Minimum inven...
14	Created Failed	I-581060		Minimum inventory must be less than the required inventory [OTM.MinLevel], 'I-581060' Minimum inven...

- If any errors are detected with the template data, the **Key** and **Description** of the failing record is shown
- Drill-down to the row in the template by clicking the **Error File** button

Depending on the error options you selected, the import run will stop when one or more errors occur.

For failed records, the window will show information to help you debug the error.

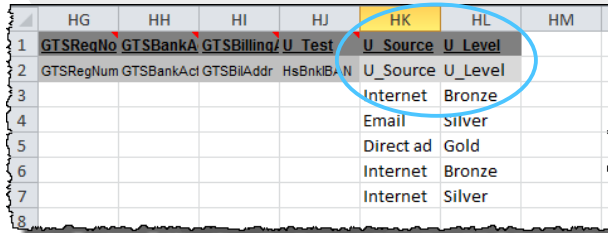
You can also drill-down to see the row in the template that failed by clicking the **Error File** button.

Working With the Templates



Importing into User-Defined Fields

- To import data into a user-defined field, add the name of the user-defined field and the data in a column at the end of the template



	HG	HH	HI	HJ	HK	HL	HM
1	GTSRegNo	GTSBankA	GTSBilling	U_Test	U_Source	U_Level	
2	GTSRegNum	GTSBankAct	GTSBilAddr	HsBnkBA	N	U_Source	U_Level
3					Internet	Bronze	
4					Email	Silver	
5					Direct ad	Gold	
6					Internet	Bronze	
7					Internet	Silver	
8							

- If an object has many user-defined fields, you have the option to generate a new template for the object that includes user-defined fields

The predefined templates obviously do not include user-defined fields.

If the object has just a few user-defined fields, you can simply add the name of the user-defined field in a column header at the end of the template and enter the data in the column.

Use the database field name for the user-defined field, which starts with “U_”.

If the object has many user-defined fields, you have the option to generate a new template for the object that includes all the user-defined fields. This is done from within the DTW application and is covered in the next section.

Adding or Updating Contact Employees for a Business Partner - 1

- You can update or add records to the child tables of an existing object without the parent template
- For example, you can add new contact employees, or update contact information, for a business partner using the child template OCPR
- Enter the key for the parent table in column A

ParentKey	LineNum	CardCode	Name	Position	Address	Phone1	Phone2	MobilePhone	E-Mail
C00001	0		Jane Briggs	MD	123 456 3458	123 456 3458		7712356901	bill@Ezimports.com
C00001			Tom Smith	Purchase Mgr	123 456 7690	123 456 7690		7089231453	jane@Ezimports.com
C00300	0		Lana Mason	Owner	345 678 7654	345 678 7654		7765213098	lana@A2.co.uk
C00300	1		Florence Ward	Buyer	345 670 9821	345 670 9821		7709214563	Flo@A2.co.uk
C00300	2			Accounts					
C00005									
C00005									

- To update an existing contact, enter the *LineNum* in column B as 0, 1, 2, etc.) to match the order in the object
- To add a new contact for a business partner, leave the *LineNum* blank

You can use the child templates without the respective parent template to update or add individual records for the child tables of an existing object. For example, you can update or add new contact employees in the child table OCPR for the business partner master data object.

In the child template, enter the code for the parent object in column A, and identify the child row using the *LineNum* field in column B. The *LineNum* field is entered as an integer: 0 for the first row, 1 for the second row, and so on.

For contact persons, the *LineNum* field is not saved in the database:

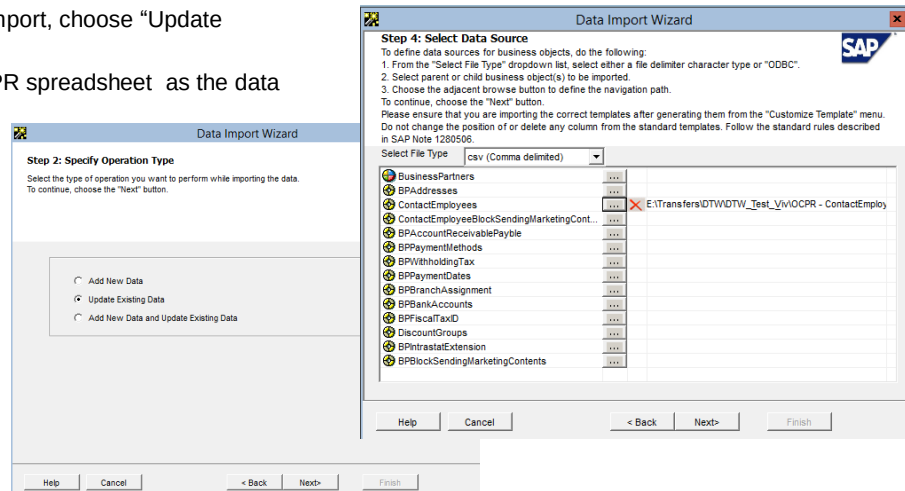
- To update an existing contact, you must enter 0 for the first contact, 1 for the second contact, and so on. The order is determined by the order the contacts were initially added and this is how they display on the Contact Persons tab of the master data record
- To add a new contact person to an existing object, leave the *LineNum* field blank

In the slide example:

- For business partner C00001, the spreadsheet will update the first contact (*LineNum* 0) and will add 2 new contacts (*LineNum* left blank)
- For business partner C00300, the spreadsheet will update 3 existing contacts (*LineNum* 0, 1 and 2)
- For business partner C00005, the spreadsheet will add 2 new contacts (*LineNum* left blank)

Adding or Updating Contact Employees for a Business Partner - 2

- When you run the import, choose “Update Existing Data”
- Only select the OCPR spreadsheet as the data source



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When you run the DTW import, choose the option to “Update Existing Data” so that the parent template is not required.

Then only select the child spreadsheet file as the data source.

Adding or Updating Addresses for a Business Partner

To add new addresses, or update existing addresses for a business partner, you do not use the LineNum field because you can identify addresses using the Address ID:

- Enter the code in column A
- In column C enter the Address ID that will be updated or added to the business partner record
- In column N enter the address type (bo_BillTo or bo_ShipTo)
- When you run the import, choose "Update Existing Data"
- Only select the CRD1 spreadsheet as the data source

The screenshot displays an Excel spreadsheet titled 'CRD1 - BPAddresses2 [Compatibility Mode] - Excel' and the SAP Business Partner Master Data interface. The Excel spreadsheet has columns: ParentKey, LineNum, AddressName, Street, Block, ZipCode, City, FederalTax, TaxCode, BuildingFlo, and AddressType. The SAP interface shows the Business Partner Master Data for C00005, with tabs for General, Contact Persons, Addresses, and Payments. The Addresses tab is active, showing 'Bill to' and 'Ship to' sections. A yellow box highlights the 'Ship Address' section in the SAP interface, and an arrow points from the 'Ship Address' row in the Excel spreadsheet to this section.

To add new addresses, or update existing billing and shipping addresses for a business partner, you do not use the LineNum field because you can identify each address using its Address ID. The Address ID can be viewed in the business partner master data. Each address for the business partner has a unique Address ID.

To update or add new billing and shipping addresses:

- Enter the code in column A
- In column C enter the Address ID to match the address that will be updated, or a new Address ID for a new billing or shipping address
- In column N enter the address type (bo_BillTo or bo_ShipTo)
- When you run the import, choose "Update Existing Data"
- Only select the CRD1 spreadsheet as the data source

In the slide example:

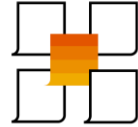
- For the customer C00001, 2 new addresses are being added
- For the customer C00005, the existing shipping and billing addresses are updated

Customized Templates



This segment covers the creation of customized templates.

Business Example for Customized Templates



- You want to import item master data into a new SAP Business One system.
- The predefined template OITM contains many fields including sales and purchasing units and properties. The products to be imported do not use many of these template fields.
- To simplify the data entry process, you can remove unused fields and generate a custom template. This will make the templates easier to work with.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	ItemCode	ItemName	ForeignName	ItemsGroup	CustomsCode	SalesVATCode	BarCode	VATLabel	PurchaseSet	SalesItem	InventoryItem	IncomeAccount	ExemptCode	ExpenseAccount	MainSupplier	SupplierCode	DesiredInventory	MinInventory	Picture	UserText	SerialNum	Commission	Commission
2	ItemCode	ItemName	ForeignName	ItemsGroupCode	CustomsCode	SalesVATCode	BarCode	VATLabel	PurchaseSet	SalesItem	InventoryItem	IncomeAccount	ExemptCode	ExpenseAccount	CardCode	SupplierCode	ReorderQty	MinLevel	PictureName	UserText	SerialNum	CommissionPoint	CommissionSum
3																							
	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS
	CommissionSum	CommissionGroup	TreeType	AssetItem	DataExportCode	ManufactureCode	ManageSales	ManageBuy	ValidFrom	ValidTo	ValidFrom	ValidTo	ValidFrom	ValidTo	ValidFrom	ValidTo	ValidFrom	ValidTo	ValidFrom	ValidTo	ValidFrom	ValidTo	ValidFrom
	AT	AU	AV	AW	AX	AY	AZ	BA	BB	BC	BD	BE	BF	BG	BH	BI	BJ	BK	BL	BM	BN	BO	BP
	SalesUnit	SalesUnit	SalesUnit	SalesUnit	SalesUnit	SalesUnit	SalesUnit	SalesUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit	PurchaseUnit
	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit	SerialUnit

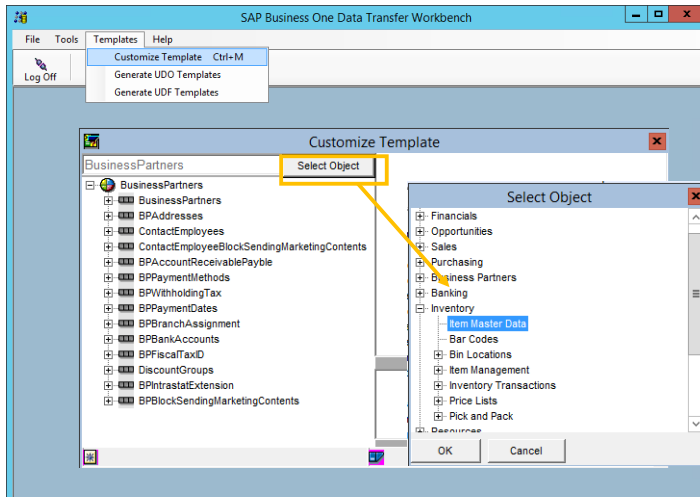
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Some objects in SAP Business One contain many, many fields. Examples are the business partner master data (OICRD) and the item master data (OITM).

In this example, the products being imported do not use many of the fields in the OITM template. To simplify the process of entering data in the template, you can remove unused fields and generate a custom template.

Customize Template - 1



- To create a template tailored to your needs, choose *Templates>Customize Template*
- Select the business object which will be the basis for the new template

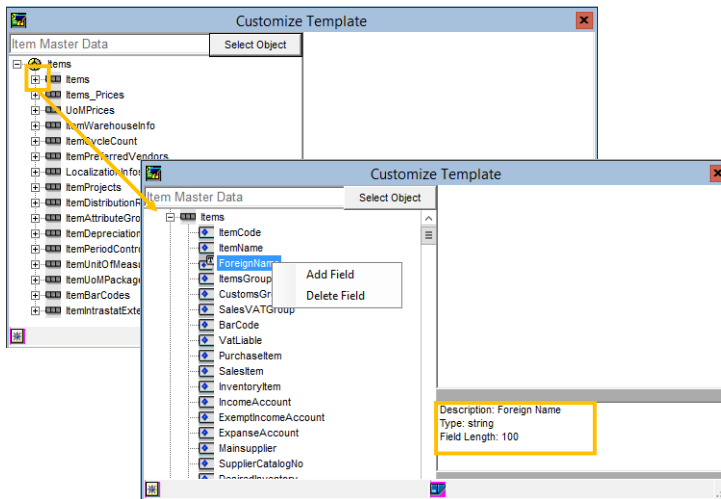
You may have noticed that the predefined templates are based on a preconfigured data transfer interface that maps to SAP Business One tables.

You can create custom templates to better match to the data you are importing. For example, if you do not use most of the fields in the predefined template, you can create a custom template with just the columns you need.

From the DTW user interface, select **Templates → Customize Template**.

The select the business object which will be the basis for the new template. In the example the item master data object is selected.

Customize Template - 2



- Expand the node for the table to see the complete list of DI API fields
- When you select a field, the field's attributes display in the lower right
- You can:
 - Rearrange the field order by dragging and dropping
 - Delete a field from the template
 - Add a previously removed field

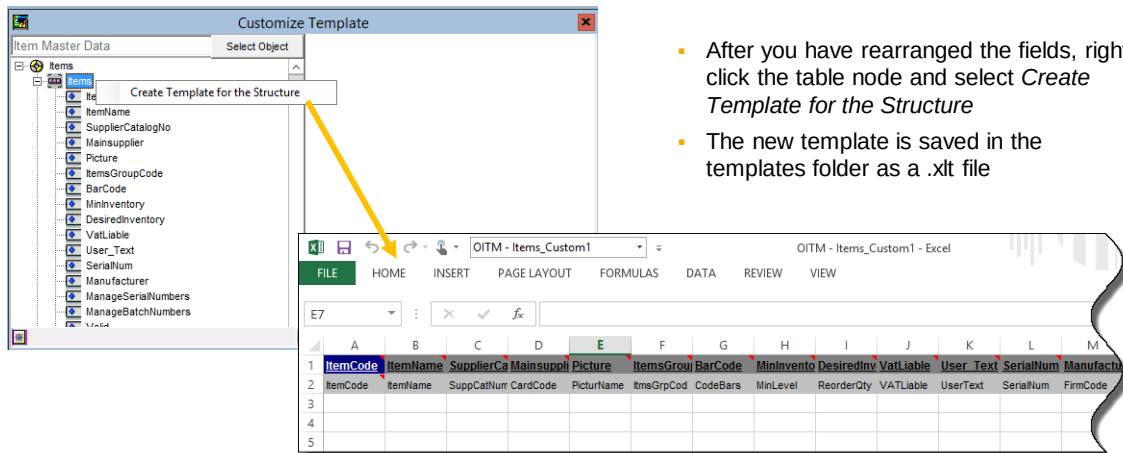
When you expand the node for an object, such as the Items object, you see a list of the fields from the predefined template. When you select a field you can see the field's attributes displayed in the lower right.

You can rearrange the order of the fields by dragging and dropping. So you can move the fields you use the most to appear first in the template.

You can also right-click on a field and either remove it from the template, or add back a previously deleted field. Do not remove mandatory fields or primary keys from the template.

You can only move or delete one field at a time.

Customize Template - 3

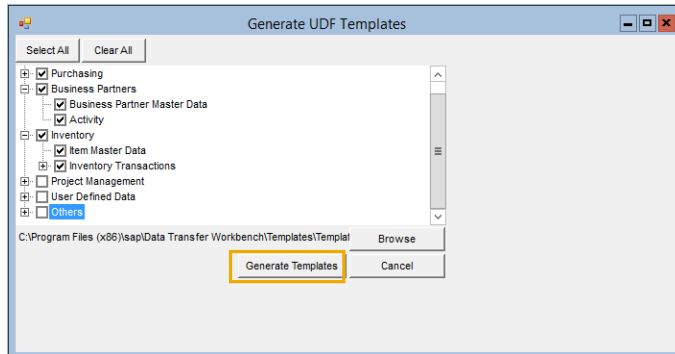
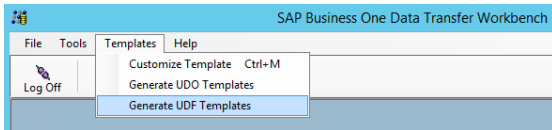


The screenshot shows the 'Customize Template' dialog box on the left, which is part of the SAP system. It displays a tree view of the 'Item Master Data' structure. A yellow arrow points from the 'Create Template for the Structure' button in the dialog to the 'Items' table node in the tree. On the right, an Excel spreadsheet titled 'OITM - Items_Custom1 - Excel' is shown. The spreadsheet has a header row with the following columns: ItemCode, ItemName, SupplierCa, Mainsuppli, Picture, ItemsGrou, BarCode, MinInvento, DesiredInv, VatLiab, User_Text, SerialNum, and Manufactu. The data rows are empty, and the spreadsheet is saved as a .xlt file.

- After you have rearranged the fields, right-click the table node and select *Create Template for the Structure*
- The new template is saved in the templates folder as a .xlt file

After you have rearranged the fields, right-click the table node and choose the option to create a new template. The template is saved as an .xlt file. You can then enter your data into the template and save as a delimited file for the import.

Generating UDF Templates



User-defined fields (UDFs) added to an object do not appear in the predefined templates, but you can generate a new template to include the UDFs:

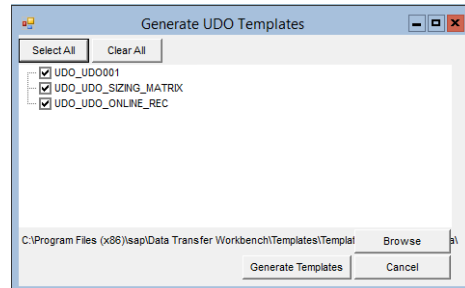
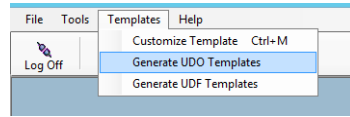
- Choose *Generate UDF Templates*
- The window shows a list of all objects with UDFs
- The system will generate templates for selected objects with the UDFs included in the template
- The templates are saved in the file location shown in the window

- User-defined fields (UDFs) added to an object do not appear by default in the predefined templates. If there are only a few UDFs for the object, you can add them manually at the end of the predefined template. If there are many UDFs for an object you can generate a new template that includes the UDFs.
- Choose the *Generate UDF Templates* option from the *Templates* menu. A window opens showing all objects that contain UDFs. Deselect objects for which you do not want templates generated.
- When you choose *Generate Templates* the system generates a new template for each of the selected objects with the UDFs. Included in the template. Columns for the UDFs are automatically added to the end of the predefined template.
- The new templates are saved in the location displayed in the window. You can browse to select a different location.

Generating UDO Templates

You can also generate new templates for user-defined objects (UDOs):

- Choose *Generate UDO Templates*
- The window shows all user-defined objects that have been registered (type Master Data or Document)
- The system will generate templates for selected user-defined objects



User-defined Objects (UDOs) are tables added to the database by the partner or by a user.

Often it is easier to populate these objects using DTW.

You can generate DTW templates for the UDO tables by choosing *Generate UDO Templates* from the *Templates* menu.

The window will show all User-defined Objects that have been registered with the database (object type is Master Data or Document).

A template will be generated for each selected UDO.

Summary



Key points from this course:

- Data Transfer Workbench (DTW) makes it possible to import setup data, master data, and open transactions in bulk.
- DTW provides predefined templates for the objects in SAP Business One, based on the DI API interface. The Database Tables Reference provides documentation for the templates.
- When you use a SAP HANA database you can increasing performance by checking the *Enable Faster Import* checkbox. DTW will import the data using multiple parallel threads.
- After you have entered the data in the template, save the templates as a tab, comma, or semi-colon delimited file before you run the import wizard. SAP recommends that you run a simulation first, or import the data into a test database so you can validate the results.
- You can create customized templates; for example, a template with only the columns you need for the data to be imported.
- You can also generate new templates for an object that include user-defined fields (UDFs). And you can generate new templates for registered user-defined objects (UDOs).

Here are some key points to take away from this course. Please take a minute to review these key points:

- The Data Transfer Workbench (DTW) makes it possible to import setup data, master data, and open transactions in bulk.
- DTW provides predefined templates for the objects in SAP Business One, based on the DI API interface. The Database Tables Reference provides documentation for the fields in the templates, including default values for fields, as well as field lengths and constraints on the data.
- When you use a SAP HANA database you can increasing the performance of the import by checking the *Enable Faster Import* checkbox . DTW will import the data using multiple parallel threads.
- After you have entered the data in the template, save the templates as a tab, comma, or semi-colon delimited file before you run the import wizard. SAP recommends that you run a simulation first, or import the data into a test database so you can validate the results.
- You can create customized templates for ease of use. For example, if you do not use most of the columns in the predefined template, you can create a customized template with only the columns you need to import.
- You can also generate new templates for an object that include any user-defined fields (UDFs) added to the object. And you can generate new templates for registered user-defined objects (UDOs) so you can import data into them using DTW.

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